Driver Performance Prediction

Advanced Machine Learning for Logistics

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Data Science

Problem & Data Summary

- Lack of objective data on driver safety habits.
- Identifying "High-Risk" behaviors before accidents occur.
- Manual logs are unreliable and non-scalable.

- Source: Telemetry.csv (Kaggle).
- Cleaning: Handling sensor noise and calculating time-deltas.

Feature Engineering Highlights

From raw GPS pings to behavioral metrics:

Calculated counts for Hard Brakes and Harsh Acceleration per trip.

Extracted Trip Duration, Max Speed, and Idling Time averages.

Engineered Hour of Day and Day of Week to capture temporal risk.

Model Performance Metrics

Training Results using Random Forest Regressor:

1.00 R-Squared Accuracy

1.13 Mean Absolute Error (MAE)

0.0068 Root Mean Squared Error (RMSE)

SHAP Insights and Recommendations

Top Risk Drivers

- Hard Brakes: The #1 predictor of a poor safety score.
- Harsh Accel: Secondary indicator of aggressive driving.
- Max Speed: High correlation with decreased stability.

Business Actions

- Focus training specifically on braking habits.
- Reward top-scoring drivers to improve retention.
- Predict brake wear based on harsh event frequency.

Next Steps & Future Integration

- **Fleet Dashboard**

  Visualize scores in a management portal using PowerBI or Tableau.

- **Real-Time Streaming**

Advanced Features

- Incorporate Weather/Traffic data.
- Online Learning: Re-train as drivers adapt.
- Edge Deployment: Run scoring on OBD-II devices.